SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Class : Second Year (SY) 2026-27 **Department : Information Technology**

Course code: ITXS37 **Subject: DSA**

Experiment No:3

Aim: Experiment No. 03 infix to postfix

Program

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#define MAX 100
char stack[MAX];
int top = -1;
void push(char c)
{
   if (top == MAX - 1) {
      printf("Stack Overflow\n");
      exit(1);
   }
```

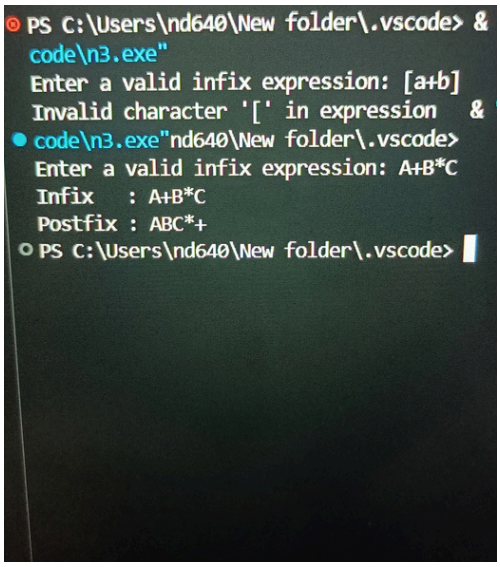
```
    stack[++top] = c;
}
char pop(void)
{
    if (top == -1) {
        printf("Stack Underflow (invalid expression)\n");
        exit(1);
    }
   return stack[top--];
}
char peek(void)
{
    return (top == -1) ? '\0' : stack[top];
}
int isEmpty(void)
{
    return top == -1;
}
int precedence(char op)
{
    switch (op) {
        case '^': return 3;
        case '*':
        case '/':
        case '%': return 2;
```

```
      case '+':
       case '-': return 1;
       default : return 0;
   }
}
int isOperator(char c)
{
   return (c == '+' || c == '-' || c == '*' ||
          c == '/' || c == '%' || c == '^');
}
int isRightAssociative(char c)
{
   return c == '^';
}
void infixToPostfix(const char infix[], char postfix[])
{
   int i, k = 0;
   char c;
   for (i = 0; infix[i] != '\0'; i++) {
      c = infix[i];
      if (isspace((unsigned char)c))
         continue;
      if (isalnum((unsigned char)c)) {
         postfix[k++] = c;
      }
```

```
else if (c == '(') {
    push(c);
}
else if (c == ')') {
    while (!isEmpty() && peek() != '('
        postfix[k++] = pop();
    if (isEmpty()) {
        printf("Mismatched parentheses\n");
        exit(1);
    }
    pop();
}
else if (isOperator(c)) {
    while (!isEmpty() && peek() != '(' &&
          (precedence(peek()) > precedence(c) ||
          (precedence(peek()) == precedence(c) &&
           !isRightAssociative(c)))) {
        postfix[k++] = pop();
    }
    push(c);
}
else {
    printf("Invalid character '%c' in expression\n", c);
    exit(1);
}
```

```
    }
    while (!isEmpty()) {
        c = pop();
        if (c == '(') {
            printf("Mismatched parentheses\n");
           exit(1);
        }
        postfix[k++] = c;
    }
    postfix[k] = '\0';
}
int main(void)
{
    char infix[MAX], postfix[MAX];
    printf("Enter a valid infix expression: ");
    fgets(infix, sizeof(infix), stdin);
    infix[strcspn(infix, "\n")] = '\0';
    infixToPostfix(infix, postfix)
    printf("Infix   : %s\n", infix);
    printf("Postfix : %s\n", postfix);
   return 0;
}
```

Outcome:


PS C:\Users\nd640\New folder\.vscode> &
code\n3.exe"
Enter a valid infix expression: [a+b]
Invalid character '[' in expression
code\n3.exe"nd640\New folder\.vscode>
Enter a valid infix expression: A+B*C
Infix : A+B*C
Postfix : ABC*+
PS C:\Users\nd640\New folder\.vscode>

Conclusion: SUCESSFULLY EXCUTED USING STRING